

■ Cravyo Escrow Payment System

Architecture & Technical Specification Document for Engineering & Product Team

1. Executive Summary

Cravyo is a peer-to-peer (P2P) food marketplace connecting **Hostelers** (buyers) with **Dayscholars** (home cooks) within college campuses. To eliminate food fraud, build 100% mutual trust between students, and automate platform revenue, Cravyo will implement an **Escrow Payment System** powered by **Razorpay Route** (India) and **Stripe Connect** (Global).

Core Rule: Money paid by a Hosteler is *never* sent directly to the cook upfront. Funds are locked securely in Cravyo's Escrow Vault and released to the cook only after dual-proof delivery and 4-digit OTP verification.

2. High-Level Escrow Order Lifecycle

Phase	Trigger Event	Order Status	Escrow Vault Action	Money Location
1. Order Created	Hosteler clicks 'Order Now'	Pending	Payment session initialized	Hosteler Account
2. Payment Locked	Hosteler completes UPI pay	Payment_In_Escrow	Hold transfer generated (on_hold: true)	Cravyo Escrow Vault
3. Cooking Proof	Cook uploads kitchen photo	Preparing	Funds remain locked	Cravyo Escrow Vault
4. Out for Delivery	Cook starts hostel delivery	Out_For_Delivery	Funds remain locked	Cravyo Escrow Vault
5. OTP Verified	Cook inputs 4-digit OTP	Delivered	Escrow Hold Released (on_hold: false)	90% Cook / 10% Cravyo
6. Auto-Refund	Cook cancels or 30m timeout	Cancelled	Full refund initiated	Refunded to Hosteler

3. Financial Breakdown & Revenue Calculator

Here is how revenue and gateway fees split automatically on a ■100 Order:

Item	Amount (■)	Percentage / Details
Total Charged to Hosteler	■100.00	100% Customer Payment
Payment Gateway Fee (Razorpay 2% + 18% GST)	-■2.36	Covered by Cravyo's Fee
Dayscholar Cook Direct Payout	-■90.00	90% Base Food Price
NET CRAVYO PLATFORM PROFIT	+■7.64	7.64% Net Commission Profit

4. Database Schema Extensions (MongoDB Mongoose)

Order.js Schema Additions:

```
paymentStatus: {
  type: String,
  enum: ["Pending", "In_Escrow_Hold", "Released_To_Cook", "Refunded"],
  default: "Pending"
},
razorpayOrderId: { type: String, default: "" },
razorpayTransferId: { type: String, default: "" }, // Escrow Hold ID
escrowHoldReleased: { type: Boolean, default: false },
platformFee: { type: Number, default: 0 },
cookPayoutAmount: { type: Number, default: 0 },
otp: { type: String, required: true },
isOtpVerified: { type: Boolean, default: false }
```

User.js Schema Additions (Cook Payout Account):

```
bankDetails: {
  accountNumber: { type: String, default: "" },
  ifscCode: { type: String, default: "" },
  accountHolderName: { type: String, default: "" },
  upiId: { type: String, default: "" },
  razorpayAccountId: { type: String, default: "" } // Linked Route Account
}
```

5. Security & Buyer Protection Rules

- **OTP Verification Guard:** Money is NEVER released without the Hosteler's 4-digit PIN.
- **Automatic Timeout Refund:** If a cook fails to upload cooking proof within 30 minutes, order auto-cancels with a 100% refund.
- **Dual Photo Verification:** Requires 1 Cooking Proof photo (in kitchen) + 1 Handover Proof photo (at hostel door) stored in Cloudinary.

6. Teammate Roadmap & Next Steps

1. **Founders:** Complete Razorpay Business KYC & enable 'Route' in Razorpay Dashboard.
2. **Backend Engineers:** Install `razorpay` npm package and implement `/api/orders/verify-otp-and-release-escrow` controller.
3. **Frontend Engineers:** Add Razorpay Checkout script and 'Funds Held in Escrow' trust badge on Hosteler checkout.